

## Math 290

### Practice Quiz 2 — Version C

(extra practice, Sections 6–11)

1. Short response.

- (a) Evaluate the following proposed theorem and proof.

**Theorem:** Let  $n \in \mathbb{Z}$ . If  $3n$  is even, then  $n + 7$  is odd.

*Proof.* Let  $n \in \mathbb{Z}$ . We work directly. Assume  $3n$  is even. Then  $3n = 2k$  for some  $k \in \mathbb{Z}$ , so

$$n = \frac{2k}{3} = 2 \left( \frac{k}{3} \right),$$

and  $n$  is even. Hence

$$n + 7 = 2 \left( \frac{k}{3} \right) + 7 = 2 \left( \frac{k}{3} + 3 \right) + 1,$$

which is odd. □

- A. The theorem is true and the proof is correct.
  - B. The theorem is false.
  - C. The theorem is true, but the proof is incorrect because it should have been done contrapositively.
  - D. The theorem is true, but the proof is incorrect because it assumes that  $3n$  is even.
  - E. The theorem is true, but the proof is incorrect because it claims that  $k/3$  is an integer without justifying it.
- (b) True or false: The statement “Let  $x \in \mathbb{R}$ . If  $x < 5$ , then  $x^2 - 2x \geq -1$ .” is trivially true.
- (c) True or false: The inverse of the implication  $P \Rightarrow Q$  has the same truth table as  $P \Rightarrow Q$ .
2. Let  $x, y \in \mathbb{Z}$ . Prove that if  $x$  and  $y$  have the same parity, then  $x^2 + xy$  is even.
3. Let  $a, b, c \in \mathbb{Z}$  with  $a^2 + b^2 = c^2$ . Prove, by way of contradiction, that  $a$  is even or  $b$  is even. (You may use the fact that for every  $z \in \mathbb{Z}$ , either  $z^2 \equiv 0 \pmod{4}$  or  $z^2 \equiv 1 \pmod{4}$ .)
4. Prove the following statement (giving a *single* justification for every step you make). For any sets  $A$ ,  $B$ , and  $C$ , we have  $A - (B \cap C) = (A - B) \cup (A - C)$ .